

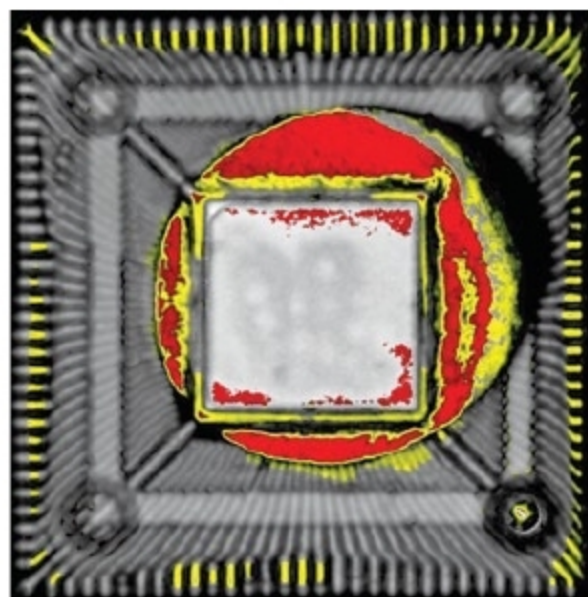


Fig. 1: C-mode top-view image of a plastic-encapsulated IC with a popcorn crack (circular feature)



Fig. 2: Q-BAM non-destructive vertical slice through the IC package along the white line shows curvature of the crack

als at the interface. But if one of the interfaces is not a solid but a gas such as air, amplitude is extremely high—close to 100 per cent of the ultrasound is reflected as a very high amplitude echo. This is what makes it so easy for a C-SAM tool making horizontal images to find cracks, voids and other gaps, and to ensure that non-gap interfaces are intact and in the right location.

Echoes returning from different depths within the sample will arrive at the transducer at different times. This allows the system to assign a physical depth to the material interface

that returned each echo. In most horizontal imaging, echoes are collected only from a depth (known as a gate) of highest interest, while echoes from above and below this depth are ignored. Gating on only part of the sample's thickness avoids having two features in the same x-y location but at different depths superimposed in the acoustic image.

Q-BAM imaging

In Q-BAM imaging the transducer scans back and forth only along the single line defined by the operator. As in C-mode imaging, it sends thousands of pulses into the sample each second and receives return echoes where there are material interfaces. On the first pass, only the echoes from a very thin gate at the bottom of the sample are collected and used for imaging. All of the echoes above this thin gate are ignored.

At the end of this first scan along the line, the transducer moves slightly upwards to change the focus and then the transducer heads back along the same line. Once again, echoes are returned from material interfaces at all depths, but only echoes from this second very thin gate will be collected to become pixels in the cross-sectional acoustic image. After numerous

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